



Group Members

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Guide Name

Dr. A. S. Mokhade sir

Practical Title: CSP529 Technical Writing & Publishing : Can AI Write Your Technical Report?

Problem Statement: Develop a Drone-based Crop Disease Detection System.

Stage 1 & 2 : Problem Definition - Student Version

Problems which led to development of Drone Based Crop Disease Detection

System:-

- Problem/Issue is that the traditional method of looking for crop disease by covering a large area takes a lot of human effort and requires a lot of time, and even after all the effort it is easier to miss the infected patch of crop when inspecting by eyes.

Reason for existence of the problem:-

- The reason for existence of the problem arises from the fact that of human physical limitation which is, small area of observation and perceiving the information incorrectly as it is hard to keep a notice of all information for the human mind, leading them to not being able to determine the issues or spreading infection in the fields.

The stakeholders:-

- The stakeholders for the drone based CDD System would be local farmers, agriculture-based industries that uses crops as their raw materials, animal husbandry which uses husk and fodder for animal dietary.

The assumptions being made while implementing drone based CDD Systems:-

- Some of the assumptions being made while implementing the drone based CDD System would be: the system will be cheaper; efficient; able to detect disease in no time with high precision.

Information still missing before implementing the system:-

- Some of the information that would be required to implement the system are: environment condition (such as temperature; precipitation; wind speed), power supply system availability, flora and fauna around the field.

Stage 3 : AI Version

1. Problem Definition & Context

- The proposed system utilizes an autonomous unmanned aerial vehicle equipped with high-resolution optical cameras to scan agricultural fields, detect plant diseases, and optimize farm management through machine learning. **(I)**

2. Why Does the Problem Exist?

- Farmers face difficulties in crop monitoring because manual scouting across extensive acreage is time-consuming, expensive, and subject to human error. **(I)**

3. Stakeholders

- Key stakeholders include farmers, agricultural workers, drone pilots, and technology providers who want to improve agricultural yield. **(I)**

4. Assumptions

- It is assumed that the drone will have continuous battery power, weather conditions will remain clear, and high-speed internet will be universally available. **(A)**

5. Missing Information

- Missing data regarding exact battery mAh capacity limits, payload weight restrictions, and offline data buffering protocols when cellular telemetry fails. **(I)**

Stage 4 : Comparison Table: Human vs. AI Performance

Evaluation Metric	AI Output	Student Output
Problem Definition	Autonomous UAV with optical cameras and ML for disease detection and farm optimization	Manual visual inspection is time-consuming, labour-intensive, and prone to missing infected patches
Reason for Existence	Manual scouting across large areas is time-consuming, expensive, and error-prone.	Human physical limitations restrict observation areas and lead to missed or incorrectly perceived infections.
Stakeholders	Farmers, agricultural workers, drone pilots, and technology providers.	Local farmers, agriculture-based industries, and animal husbandry.
Assumptions	Assumes continuous battery power, clear weather, and universal high-speed internet.	Assumes the system will be cheaper, efficient, and highly precise.
Missing Information	Exact battery capacity, payload weight restrictions, and offline data buffering protocols.	Environmental conditions (temperature, wind, etc.), power supply availability, and surrounding flora/fauna.

Stage 5: Cross-Group Insights

1. What practical difficulties will arise?

Some of the practical difficulties that will arise include legacy infrastructure integration for the technology, maintaining consistent power and connectivity, and dealing with sensor drift due to pressure and corrosion over a period of time.

2. What data will be unavailable?

Some of the information that will be unavailable to the system includes historical leakage records, real-time pipeline conditions, and precise flow rates.

3. What ethical issues exist?

The ethical issues that exist while implementing this system involve algorithmic bias, unwanted surveillance, and establishing clear accountability for system failures.

4. What assumptions are unrealistic?

Some of the assumptions that are unrealistic are expecting perfect accuracy from the system and assuming that real demands can be definitively determined using only previous trends.

5. What happens if sensors fail?

If the sensors fail, the system will need to immediately trigger fail-safe protocols and rely on secondary verification logic to prevent catastrophic failure.

6. What happens during floods?

During severe environmental events like floods, the system faces a severe hardware shortage risk and a high contamination risk.

7. What happens when the internet is unavailable?

When the internet is unavailable, cloud computing fails, which makes it strictly necessary for the system to fall back and operate in a localized offline mode.

Stage 6: Revised Technical Document

1. What is the problem?

- Manual ground scouting across a 100-hectare plot requires 40 man-hours per inspection cycle, resulting in an estimated 7% miss rate of early-stage foliar necrosis due to visual occlusion.

2. Why does it exist?

- Human ocular limitations restrict the effective field-of-view to $< 2\%$ ground cover per visual sweep and induce cognitive fatigue over extended acreages, preventing accurate pre-symptomatic identification of fungal hyphae.

3. Who are the stakeholders?

- Primary stakeholders include local tenant farmers managing seasonal crop yields, commercial grain processing facilities requiring uniform raw gluten content, and livestock feed suppliers dependent on uncontaminated crop residue (husk and fodder).

4. What assumptions are being made?

- System deployment assumes a capital expenditure recovery within 2 harvest cycles, an automated processing time under 45 minutes per 100 hectares, and a spectral classification accuracy threshold exceeding 92%.

5. What information is still missing?

- Pre-deployment metrics lack real-time microclimate thresholds (wind shear velocity exceeding , ambient temperature fluctuations), local battery charging infrastructure specifications, and localized baseline reflectance spectra for native weed flora.

Stage 7: What did AI write that looked convincing but was actually weak?

What AI wrote:

The AI stated that the system "utilizes an autonomous unmanned aerial vehicle equipped with high-resolution optical cameras to scan agricultural fields, detect plant diseases, and optimize farm management through machine learning". It also explicitly assumed that "the drone will have continuous battery power, weather conditions will remain clear, and high-speed internet will be universally available".

Why it looked convincing:

The text was grammatically flawless and heavily utilized authoritative, professional buzzwords (e.g., "autonomous," "machine learning," "optimize"). It presented a seamless, idealized solution that reads exactly like a polished technology pitch or a high-level executive summary.

Why it was actually weak (Engineering Flaws):

- **Ignored Physical Constraints:** Assuming "continuous battery power" completely dismisses the fundamental engineering trade-offs between a drone's payload weight (sensors/cameras) and its rapid battery discharge rate.
- **Disconnected from Field Reality:** Assuming "clear weather" and "universal high-speed internet" demonstrates a severe lack of domain knowledge regarding unpredictable rural microclimates and the frequent cellular telemetry dropouts in agricultural zones.
- **Zero Measurability:** Phrases like "optimize farm management" are generic filler. They provide no specific metrics, operational parameters, or quantifiable thresholds that a real engineer would need to actually build or evaluate the system.

Final Reflection

- **Student 1 : Gaurav Acharya MT26MCS024**

Three things AI cannot do for me while writing technical documents:

- **Conduct physical reality checks:** AI cannot take a drone into a field to test how morning dew actually affects the optical camera lenses.
- **Calculate real-world degradation:** It cannot measure the exact battery drain caused by sudden wind shear or the long-term corrosion of sensors due to humidity.
- **Gather human intelligence:** AI cannot interview a local farmer to find out if they actually have the technical literacy to operate the drone's interface.

Three things AI can do better than me:

- **Formatting heavy data:** AI can instantly format pages of raw, messy telemetry data into clean, readable comparison tables.
- **Synthesizing manuals:** It can rapidly summarize a 200-page civil aviation regulatory manual into a concise list of payload restrictions.
- **Syntax correction:** It perfectly spots passive voice and rephrases my hardware architecture descriptions into active, precise engineering language.

What skills must I now develop as a researcher?

- I must develop **field-testing validation and observational rigor**. I need to be the bridge between theoretical AI text and actual physical constraints, ensuring that every claim I write can survive contact with the real world.

Final Reflection

- **Student 2 : Prasanna Thulkar MT26MCS028**

Three things AI cannot do for me while writing technical documents:

- Audit data integrity: AI cannot verify if the historical crop disease data used to train our models contains regional biases or mislabelled variables.
- Set acceptable risk margins: It cannot decide the ethical and operational threshold for a "false positive" (e.g., whether it is better to over-spray or under-spray a field).
- Identify unknown unknowns: AI hallucinates solutions when data is missing, rather than intelligently recognizing and documenting what specific algorithmic logic is unavailable.

Three things AI can do better than me:

- Drafting boilerplate logic: AI is exceptionally fast at writing standard definitions for multispectral indices (like NDVI or NDRE) and basic algorithmic workflows.
- Maintaining structural flow: It can quickly organize a chaotic list of software requirements into a perfectly structured, hierarchical document.
- Translating code to text: It can seamlessly translate blocks of Python data-processing scripts into plain, readable English for the methodology section.

What skills must I now develop as a researcher?

- I must develop algorithmic accountability and critical data evaluation. My job is no longer just writing code or text, but aggressively questioning the assumptions, biases, and edge cases hidden within the systems I am documenting.

Final Reflection

- **Student 3 : Sandeep MT26MCS032**

Three things AI cannot do for me while writing technical documents:

- **Evaluate economic viability:** AI cannot accurately determine the real-world willingness-to-pay of local tenant farmers navigating tight seasonal budget margins.
- **Make strategic trade-offs:** It cannot decide whether to sacrifice a sensor's detection accuracy by 5% to save 30% on manufacturing costs to make the product viable for developing markets.
- **Navigate human ethics:** AI cannot manage the privacy concerns or legal liabilities associated with flying cameras over privately owned agricultural land.

Three things AI can do better than me:

- **Tailoring communication:** AI can take one highly technical paragraph and instantly rewrite it three different ways (for an engineer, an investor, and a farmer).
- **Drafting executive summaries:** It can extract the core value propositions from a 20-page technical report and condense it into a perfect half-page summary.
- **Brainstorming risk matrices:** It is excellent at rapidly generating a broad list of potential generic business and operational risks to serve as a starting point for discussion.

What skills must I now develop as a researcher?

- I must master strategic cross-functional analysis. I need the ability to look at a technical report and immediately connect the engineering constraints with business viability, ethical compliance, and actual stakeholder needs.